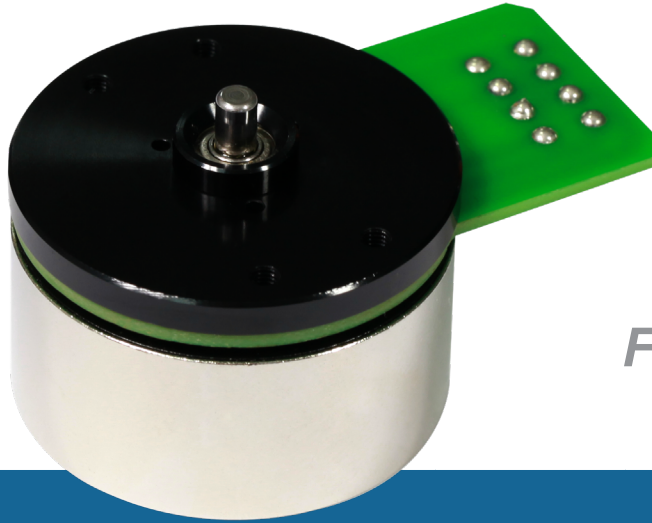




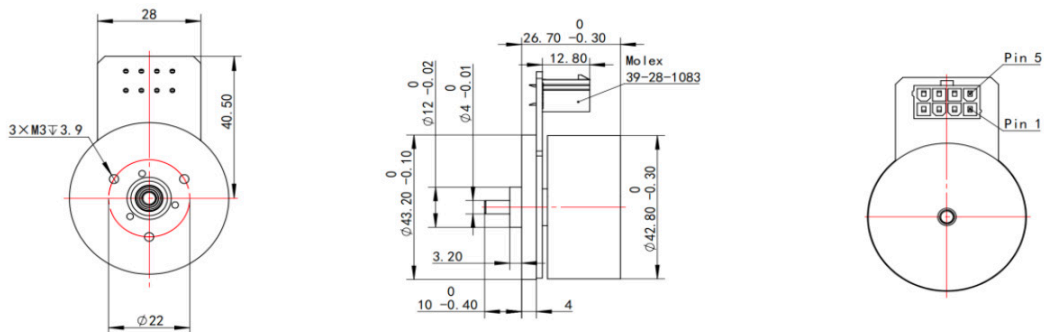
EC4527

FLAT BRUSHLESS DC MOTOR, 70 WATT

Motor Data				
Values at Nominal Voltage	Part Number 200011	Part Number 200012	Part Number 200013	Part Number 200014
Nominal voltage, V	24	30	36	48
No load speed, rpm	6,110	6,230	6,330	3,440
No load current, mA	400	253	229	112
Stall torque, mNm	1,150.0	1,040.0	1,000.0	897.0
Stall current, A	39.5	25.8	20.7	7.0
Max. efficiency, percent	85	84	83	84
Nominal power, W	62.2	55.9	54.9	34.0
Nominal speed, rpm	4,860	4,990	5,080	2,540
Max. continuous torque, mNm	128	112	108	134
Max. continuous current, A	3.21	2.36	1.93	0.94
Characteristics				
Terminal resistance phase to phase, Ω	0.55	1.16	1.74	6.89
Terminal inductance phase to phase, mH	0.22	0.32	0.53	3.12
Back-EMF constant, mV/rpm	3.89	4.77	5.62	13.73
Torque constant, mNm/A	37.17	45.53	53.71	131.10
Speed constant, rpm/V	257	210	178	73
Current constant, A/mNm	0.027	0.022	0.019	0.008
Speed / torque gradient, rpm/mNm	5.31	5.99	6.33	3.84
Mechanical time constant, ms	8.1	10.3	11.1	7.2
Rotor inertia, gcm ²	181	181	181	181

Motor Data				EC4527
Specifications	Part Number 200011	Part Number 200012	Part Number 200013	Part Number 200014
Operating temperature range:				
Motor, °C	-40 — 100			
Winding, max permissible, °C	125			
Direction of rotation	electronically reversible			
Max. speed, <i>rpm</i>	10,000			
Number of pole pairs	8			
Number of phases	3			
Magnet material	NdFeB			
Hall sensors	digital			
Mechanical Data				
Bearing type	ball bearings, preloaded			
Axial play, <i>mm</i>	0 — 0.3			
Max. axial load (dynamic), <i>N</i>	3.8			
Max. force for press fits (static), <i>N</i>	50			
(static, shaft supported)	1,000			
Max. radial loading, 5 mm from flange, <i>N</i>	21			
Product				
Weight of motor, <i>g</i>	141			

DIMENSIONAL DRAWING



OPTION, CABLE AND CONNECTION INFORMATION

Connector: Molex 39-28-1083		Connector: Molex 39-28-1083	
Hall sensors A	Pin 1	Hall sensors C	Pin 5
Hall sensors B	Pin 2	GND	Pin 6
V _{hall} 4.5V...24V	Pin 3	Motor winding A	Pin 7
Motor winding C	Pin 4	Motor winding B	Pin 8